

troubleshooting scenarios such as the current one. Protocol Hierarchy analyses captured files using various parameters beginning with Ethernet – IPv4 / IPv6, and drilling down TCP and UDP to corresponding details—such as http, ssl, pop for TCP and DNS, etc. for UDP. These files are presented as % bytes used by these protocols with respect to total traffic. Look at Figure 3 for an interesting capture showing too much IPv6 traffic. (Later, it was concluded that the traffic was LAN file transfer traffic.)

The traffic was further analysed to isolate TCP Keep-Alive, TCP Retransmission, RST-ACK and DUP-ACK packets, which generally signify problems. Systems found with these packets (faulty systems) were identified and isolated from the network. The systems were also scanned for viruses, and the infected systems were removed. However, with no further luck here too, troubleshooting had to continue.

Look carefully at Figure 4. To isolate RST and ACK traffic, the display filter `tcp.flags == 0x0014` was used.

This was really frustrating and additional time was wasted with the trial and error process. Yet, the problem was still unresolved, and customer anxiety was increasing day by day.

Going back to the basics, it occurred to us that the issue could be with the NIC drivers or operating systems, primarily due to the fact that the problem started only after the new computer systems were installed. So, a few old systems that were using the same NICs were checked, which revealed a driver version that was different than the OS default drivers and the current drivers available on the vendor's website (which were applied in Step 1). Matching the old drivers on all the systems finally solved the issue.



Figure 3: Protocol Hierarchy

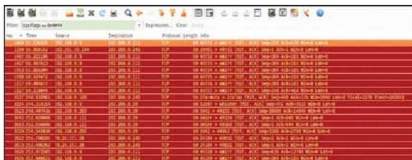


Figure 4: RST and ACK packets



Warning

Packets captured using the test scenarios described in this series of articles are capable of revealing sensitive information such as login names and passwords. Using ARP spoofing, in particular, will disturb the network temporarily. Make sure to use these techniques only in a test environment. If at all you wish to use them in a live environment, do not forget to avail explicit written permission before doing so.

Looking back

Though packets indicating network issues were present in the capture, their percentage compared with good packets was just too low – which could not have affected the network to the point of bringing it down completely. It was easy to conclude from the Wireshark capture that the top down approach starting from the operating system level and going down towards the physical layer should have been adhered to before going ahead with removing the supposed 'faulty systems' from the network.

The end user was appreciated for his patience and confidence in the troubleshooter.

For troubleshooters, 'slow network' issues are indeed a challenging task. Apart from resourceful troubleshooters with adequate tools, techniques, experience and insight, successful troubleshooting also requires a lot of patience from the engineer and the customer as well! **END**

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